

CS2430 Project #1 Design notes/decisions
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- Data structure:
 - Utilize an ordinary array data structure for every algorithm; `int[]`
- Invariants:
 - The integers in each array do not change (no duplicating/deleting/replacing)
 - Each sorting algorithm sorts the same arrays
- Assumptions:
 - Array size > 1
 - Inputs are small integer arrays
 - Comparison count = 0 at the start of each sort
- Tradeoffs:
 - Simplicity vs. performance (avoiding $O(n^2)$ complexity where possible)
 - Making a copy of an array for each sorting algorithm takes more storage but ensures that no algorithm is sorting another algorithm's output